Evaluating AI Judges of Formal Specification Coverage with Conditional Human-Rating Distributions

Anonymous submission

Abstract

Artificial intelligence (AI) judges are useful evaluation instruments only to the extent that their ratings measure the intended property. For formal specification coverage, a high correlation with experts does not establish agreement about score levels or disagreement between experts on individual artifacts. We specify a study in which a judge predicts the six-category human-rating distribution conditional on Python source, a candidate Lean formalization, and a gold reference. The primary objective is normalized ranked probability score, with correlations and absolute error retained as diagnostics. A prospective design covers 896 benchmark tasks and four solver configurations, with 24 distinct experts and a starting budget of 14,848 individual ratings. Separate task, solver, and expert holdouts test different transfer claims, while controlled specification changes test whether human agreement reflects the intended construct. Historical VeriBench evidence motivates the design: a 30-artifact, ten-task test has Spearman correlation 0.826, but its 95% task-bootstrap interval is [0.316, 0.966] and its normalized judge-minus-expert mean bias is +0.181. Those previously exposed data do not validate the proposed judge; fresh human collection and the prospective results remain pending. The contribution is a reproducible measurement protocol that separates predictive agreement, semantic validity, and uncertainty about both.

1 Introduction

AI agents increasingly produce artifacts whose quality cannot be reduced to whether their code executes. A formalization agent, for example, can produce a Lean program and prove every theorem it states while omitting a required property of the original Python program. An evaluation instrument must distinguish successful proof construction from adequate specification of the task. This distinction matters when scientific agents use formal artifacts as evidence: correctness of a weak claim does not establish completeness of the intended claim.

Large language models (LLMs) can compare candidate artifacts with instructions and references at a scale that expert review cannot easily match. Prior work on LLM judges demonstrates useful human agreement while identifying position, verbosity, and self-preference effects (Zheng et al. 2023). In the present setting, the immediate target is formal specification coverage (FSC): whether candidate theorem statements express the properties required by the task, potentially through logical implication rather than textual similarity. Human ratings provide an operational comparison for this semantic assessment.

Correlation alone leaves the central question unresolved. A judge can preserve the ordering of artifacts while systematically assigning higher scores than experts. Two judges can also have identical pooled score distributions while assigning those scores to different artifacts. Furthermore, averaging all expert ratings into one target erases whether a particular artifact produces consensus or disagreement. These are distinct failures, and collecting more ratings does not repair the choice of an unsuitable objective.

We therefore define the target as a conditional distribution over six ordinal human ratings and evaluate probability forecasts with a proper scoring rule. The judge receives the source program, candidate formalization, and reference, together with a fixed rubric. It predicts how qualified experts following that rubric would distribute their ratings. Normalized ranked probability score (RPS) gives an explicit objective for this target; correlations remain useful summaries of expected scores, rather than a substitute for distributional agreement.

This paper specifies the corresponding data collection, selection, and evaluation protocol. It distinguishes the number of tasks, the number of solver configurations, the total number of human raters, and the number of ratings assigned to an artifact. The starting design covers all 896 tasks of a fixed VeriBench release and four named solver configurations, while preserving separate evidence for unseen tasks, excluded solver configurations, and unseen experts. We also preserve historical judge results with their uncertainty and exposure limitations instead of treating them as fresh validation.

The contribution is a prospective methodology and historical case study, not a completed claim that a new judge works. Proper scoring rules and dependence-aware inference are established tools; our contribution is their operational use for FSC, with auditable inputs, splits, controls, and a collection budget. The project concerns AI, agent, and LLM judges.

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A separate prover judge may use AI to search for an implication proof, but the proof checker determines whether that implication was established. Human agreement and a checked proof answer different questions, and a proof-search timeout establishes neither equivalence nor non-entailment.

2 Target and Measurement Conditions

For task t and solver configuration a, let

$$z_{ta} = (x_t, \widehat{y}_{ta}, y_t^*), \qquad R_{tah} \in \{0, 1, 2, 3, 4, 5\}. \tag{1}$$

Here x_t is the original Python source and task contract, $\widehat{y}_{ta}$ is the candidate Lean artifact, y_t^* is the reference, and h identifies a human rater. The normalized scalar rating is $R_{tah}/5$. The main study displays the complete candidate and reference artifacts to raters and the judge, including any proof bodies or `sorry` placeholders. Both receive identical input contents; the rendered view is recorded, and proof completion is a separate variable rather than the coverage target.

2.1 What coverage means

A formal specification is a theorem statement considered without its proof. Coverage concerns the required statements and the domains and assumptions under which they hold. A candidate may cover a reference statement with a stronger statement or with several statements together. Counting theorem names or measuring textual overlap cannot establish this implication. Conversely, narrowing a precondition can remove required cases even when a theorem's conclusion is unchanged. Proof completion is recorded separately from statement coverage.

The source program constrains the intended behavior but does not automatically resolve a conflict between implementation behavior, task requirements, and the reference. The protocol records such conflicts for independent inspection rather than silently choosing one input as infallible. Conditioning on a reference makes the rating task more concrete; it can also make both humans and a judge prefer the reference's representation. Reference sensitivity is therefore a measurement condition to test, not an assumed benefit of adding the reference.

2.2 The human distribution

Let $\mathcal{H}$ denote a specified population of qualified Lean and formal-methods experts using the same rubric. The target is the probability mass function

$$p^*(r \mid z) = \Pr_{h \sim \mathcal{H}}(R_h(z) = r). \tag{2}$$

This definition requires a recruitment frame and a rating procedure, including allowed assistance, time guidance, and treatment of ambiguous inputs. It does not make a convenience sample representative of all experts. The empirical distribution from a fixed recruited panel is a different, narrower target; both must be labeled when reported.

The judge predicts $q_\theta(r \mid z)$, a nonnegative vector of six probabilities summing to one. The initial interface elicits those six probabilities explicitly as structured output; it does not infer them from token likelihoods or the frequency of repeated scalar answers. Any subsequently fitted probability mapping is trained on training ratings and selected on validation ratings, with its family and constraints recorded as a separate candidate condition. Its expected normalized rating is $\mu_q(z) = \sum_{r=0}^{5}(r/5)q_\theta(r \mid z)$. Each individual human rating is retained, including disagreement. A histogram of stochastic scalar judge outputs does not by itself represent the human distribution: model-call instability and human disagreement have different sources. The proposed probability-vector interface is consequently a new judge condition, not a reinterpretation of an old scalar score.

3 A Distribution-Sensitive Objective

Proper scoring rules reward accurate probability forecasts in expectation (Gneiting and Raftery 2007). For the cumulative probabilities $Q_c = \sum_{r=0}^{c} q_r$, we use the loss

$$S(q, r) = \frac{1}{5}\sum_{c=0}^{4}\big(Q_c - \mathbf{1}\{r \le c\}\big)^2. \tag{3}$$

The factor $1/5$ places RPS on $[0, 1]$. The ordered thresholds retain the ordinal structure: probability placed many categories away from an observed rating incurs more cumulative error than probability placed in an adjacent category. Equal threshold weights are an explicit design choice, not a claim that every practical consequence of a rating is equally spaced.

Let $P_c = \Pr(R \le c \mid z)$. For each threshold, expansion of the squared error gives

$$\mathbb{E}[S(q, R) \mid z] = D(q, p^*) + U(p^*), \tag{4}$$

$$D(q, p^*) = \frac{1}{5}\sum_{c=0}^{4}(Q_c - P_c)^2, \tag{5}$$

$$U(p^*) = \frac{1}{5}\sum_{c=0}^{4} P_c(1 - P_c). \tag{6}$$

The first term measures conditional distribution error; the second is the loss remaining even for a perfect forecast of a disagreeing expert population. Because the cumulative probabilities determine all six category probabilities, the unrestricted expected-loss minimizer is uniquely $q = p^*$. This fact justifies the target of selection. It does not guarantee that a particular model class, prompt search, or finite sample will attain that target.

For equal category spacing on $[0, 1]$, the one-dimensional Wasserstein distance is

$$W_1(q, p^*) = \frac{1}{5}\sum_{c=0}^{4}|Q_c - P_c|. \tag{7}$$

Cauchy–Schwarz gives $W_1(q, p^*)^2 \le D(q, p^*)$. Thus small excess expected RPS bounds an interpretable distribution distance. More generally, for a normalized L-Lipschitz utility function f, the expectation gap is at most LW_1. This connects a distributional claim to uses such as estimating the mean rating, without requiring a separate match of every moment.

3.1 Why correlation or a pooled divergence is insufficient

Consider human scores $(0, 0.25, 0.5)$ and judge scores $(0.5, 0.75, 1)$. Pearson and rank correlations are one, although every score differs by 0.5. Alternatively, permuting predictions among artifacts preserves the marginal score histogram while destroying itemwise agreement. A divergence between pooled histograms misses that failure. The required comparison is conditional on the same artifact, then averaged across artifacts.

Logarithmic loss is another proper objective and connects to Kullback–Leibler divergence, but it ignores category order and diverges when the judge assigns zero probability to an observed category. With few ratings per artifact, directly estimating an entire human probability vector and plugging it into a divergence is also noisy. RPS can instead score each individual rating without declaring a sparse empirical histogram to be the true human distribution. We retain logarithmic loss only as an optional prespecified sensitivity analysis with a documented probability-floor policy.

3.2 Selection and artifact weighting

For n artifacts, with assigned rater set $\mathcal{H}_i$ of size m_i, define

$$\widehat{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{h \in \mathcal{H}_i} S(q_\theta(z_i), R_{ih}). \tag{8}$$

Candidate construction uses training data; selection minimizes this loss on validation data. Artifacts receiving additional diagnostic ratings retain the same total weight as other artifacts. All candidates, prompt changes, and fitted score mappings are recorded because repeated model selection can overfit a validation criterion (Cawley and Talbot 2010). Final test ratings do not enter this process.

3.3 A worked distinction between uncertainty sources

Suppose, illustratively, that half of the qualified raters assign category zero to one artifact and half assign category five. The correct human forecast places probability $1/2$ on each endpoint, so every cumulative probability P_c equals $1/2$ for $c = 0, \ldots, 4$. Its expected RPS is 0.25, entirely due to human disagreement. A judge that always assigns all probability to category zero has expected RPS 0.50: its additional distribution error is 0.25. Both forecasts can be perfectly reproducible across repeated judge calls. The deterministic endpoint forecast is therefore stable but worse at predicting the human distribution.

These numbers are an exact illustrative calculation, not observed study results. They show why a low model-repeat standard deviation cannot replace the proper-loss comparison and why a positive raw RPS cannot by itself imply a defective judge. The same distinction applies when interpreting the dense subset: additional raters estimate the human cumulative probabilities more precisely, while additional judge calls estimate the variability of the probability forecast. The two replication budgets answer different questions and must be reported separately.

Collection stage	Artifacts	Ratings
128 tasks $\times$ 4, train/validation	512	1,536
896 tasks $\times$ 4, final audit	3,584	10,752
Five extra on 128 $\times$ 4 artifacts	0	2,560
Total	4,096	14,848

Table 1: Prospective collection budget, not collected data. The two stages have separate generations and disjoint human pools. Three ratings are assigned initially to every artifact; the dense subset has eight in total.

4 Prospective Study Design

The starting population is the fixed public VeriBench release `veribench@1.3`, containing 896 Python tasks. The data source is pinned to commit `c9dfd8e` of the public Harbor dataset collection (VeriBench contributors 2026). Task contents, reference versions, instruction version, and toolchain must be hashed before collection. The four required solver configurations are Claude Code, Codex, a Gemini agent, and a Grok agent. These names define the intended panel; exact model versions, harnesses, tools, effort, and budgets remain to be frozen. They are deliberately selected configurations, not random draws from every possible future solver.

4.1 Task and human allocation

The initial allocation is 64 training tasks, 64 validation tasks, and 768 held-out tasks. Related task variants and near-duplicates are grouped before splitting. If those groups prevent the exact counts, the allocation and budget must be amended before test inspection. We recruit 24 qualified experts: eight for training and validation, and 16 different raters for the final audit. Each artifact receives three independently submitted ratings from distinct randomly assigned raters in the relevant pool. Assignment balances workload and maintains overlap between raters; it is recorded rather than inferred from a dense tensor.

The final audit covers all 896 task identities with fresh solver generations. Consequently, its 128 previously seen tasks answer a fresh-output question, while its 768 held-out tasks answer a task-transfer question. Those strata are reported separately even when a full-benchmark mean is also useful. One solver output per task–configuration pair is collected initially in each stage, without selecting the best of retries. This design does not estimate generation variance at a fixed task–configuration pair. Repeated identical bytes remain distinct generation records but do not create evidence of independent content.

We randomly select 128 of the held-out tasks before observing their ratings and collect eight total ratings for each of their four artifacts. These additional five ratings support distribution diagnostics; they do not make the empirical category histogram exact. Table 1 distinguishes new artifacts from additional ratings of an existing artifact. The resulting workload is 192 ratings per training/validation rater and 832 per audit rater under exact balancing. These are planned counts, not measured or assumed feasible time commitments.

Audit stratum	Supported question
Full audit (896 tasks)	Agreement on the complete fixed benchmark and four evaluated solver configurations.
128 seen tasks, fresh outputs	Agreement on new generations from tasks used for training or selection.
768 held-out tasks	Transfer across task groups untouched in the new selection procedure.
Four excluded-solver fits	Transfer of that procedure to each configuration excluded from its fit.
16 audit raters	Agreement with new raters, with wider population inference conditional on recruitment assumptions.

Table 2: The holdouts answer different questions. No row alone establishes accuracy on every future task, solver configuration, or rater.

4.2 What is held out

The main judge is trained and selected with all four configurations' training and validation outputs. Its audit establishes performance on the evaluated configurations, with separate new-task and new-human evidence. It cannot also establish that these same configurations were unseen by that judge.

To test transfer to each excluded configuration, we fit four additional judges. For each fit, one configuration's outputs, ratings, and prompt examples are excluded from both training and validation. The fit is evaluated on that configuration's 768 held-out tasks using audit raters. All four fits reuse the same collected audit ratings. These experiments test the transfer of the training/selection procedure, not a single judge that somehow treats all four training configurations as unseen. They also do not establish transfer to arbitrary future model versions. Historical task and solver exposure is recorded; a new split cannot erase earlier prompt design or unknown pretraining exposure.

The final human pool supplies no ratings for judge selection. An analysis conditional on the 16 recruited audit experts can establish agreement with that panel. An inference about a wider expert population additionally requires a defensible recruitment and sampling model. Holding out identities reduces direct leakage but does not by itself make either experts or solver configurations representative.

4.3 Pilot and fixed execution

A training/validation-only pilot measures rating time, disagreement, fatigue, failure rates, and variance of paired losses. It also traces one artifact through generation, extraction, human display, judge input, probability parsing, and scoring using content hashes. As an artificial pipeline control, the pilot supplies a known rating in the prompt and checks that the returned category or probability vector reproduces it through parsing and normalization. This rating-reproduction check diagnoses instruction and transport failures; it is never scored as evidence of semantic ability or included among test artifacts. Before final ratings are opened, we freeze the rubric, assignments, judge versions, analysis, repeat count, failure policy, endpoints, and practical tolerances. Task and rater exclusions, parser errors, missing outputs, and timeouts remain in the accounting. The plan is not a completed preregistration: practical thresholds and the feasibility of the proposed workload remain unresolved.

5 Precision, Selection, and Decision Rules

The three dimensions of the study address different uncertainties. More distinct tasks cover a broader set of inputs and increase the precision of task-averaged comparisons. More solver configurations probe different output distributions. More distinct raters probe variation between raters, whereas more ratings of one artifact sharpen information about that artifact's rating distribution. The product of these counts is not the number of independent observations.

5.1 A transparent planning calculation

As an illustration, suppose the human-specific contribution to normalized loss has standard deviation 0.10. For independent, representative raters, the normal-approximation 95% half-width of that contribution would be

$$1.96\frac{0.10}{\sqrt{H}} = \begin{cases} 0.080, & H = 6, \\ 0.049, & H = 16. \end{cases} \tag{9}$$

This is a hypothetical component of uncertainty, not an observed total interval. If repeated ratings share a human-specific effect, thousands of ratings by the same few raters do not remove it. The calculation motivates recruiting more distinct raters, while the actual allocation must be checked against pilot variance and workload.

Increasing ratings per artifact from three to eight reduces an independent mean's standard-error factor by $\sqrt{3/8} \approx 0.61$. That is a 39% reduction under the stated independence and common-variance approximation, not a guaranteed 39% reduction in the final study interval. Balanced reuse of raters creates cross-artifact dependence, and an exact finite-panel analysis differs from independent population sampling. These conditions are why the dense subset supplements broad task coverage instead of replacing it.

For correlations, Fisher's approximation for independent bivariate-normal pairs is

$$\mathrm{SE}(\operatorname{atanh}\widehat{\rho}) \approx 1/\sqrt{n-3}. \tag{10}$$

In the actual study, multiple outputs share a task and ratings share a rater; substituting $n = 14{,}848$ into that formula is invalid. Nor can additional data be expected to raise the true correlation: it improves estimation of whatever association exists. We therefore size the study against paired proper-loss precision and expert variation, and report correlation intervals under a dependence-aware analysis.

5.2 Estimating distribution error

The raw RPS includes the disagreement term in Equation 6. For a fixed artifact and threshold, let $X_h = \mathbf{1}\{R_h \le c\}$ and

let s_X^2 be the sample variance with denominator $m-1$. Under independent draws from the specified expert population,

$$\widehat{D}_c = (Q_c - \overline{X})^2 - \frac{s_X^2}{m} \tag{11}$$

is unbiased for $(Q_c - P_c)^2$. The subtraction removes the variance of the empirical human cumulative probability. We average across thresholds and then artifacts. Individual corrected estimates can be negative and are not clipped before aggregation. Their intervals, rather than the sign of an individual estimate, determine what distribution error remains compatible with the data.

If the estimand instead concerns a fixed panel of H raters sampled uniformly without replacement, the subtraction is $(1 - m/H)s_X^2/m$. That correction targets the finite-panel mean, not the wider population automatically. More complicated assignment and nonresponse require corresponding design-based or model-based adjustments. The dense subset makes this diagnostic more informative but cannot remove uncertainty about unrepresented experts.

5.3 Uncertainty and comparison

Crossed task and rater identities require inference that respects both sources of dependence. Methods for resampling crossed arrays motivate independent reweighting of identities under appropriate assumptions (Owen and Eckles 2012). The exact sparse-assignment procedure, coverage checks, and treatment of small rater counts must be specified using pilot data before the test is inspected. We report fixed-panel analyses separately from analyses extending to an expert population. Task resampling estimates variation over tasks like those in the benchmark; for the complete fixed benchmark it is not uncertainty caused by missing benchmark task identities. The four chosen solver configurations remain fixed in the primary analysis.

Primary comparisons use paired, artifact-weighted RPS differences between a constant training-estimated distribution, a fixed rubric/reference-assisted judge, a correlation-selected judge, and an RPS-selected judge. We also include a non-LLM theorem-count baseline to test whether a semantic model improves on a simple structural cue. Let $v = \min(1, N_{\text{candidate}}/N_{\text{reference}})$ count theorem statements without inspecting their proofs. For $u = 5v$, place probability $1 - (u - \lfloor u \rfloor)$ at $\lfloor u \rfloor$ and the remaining mass at the next category, with all mass at five when $u = 5$. This defines an ordinal probability forecast for comparison; it does not assert that counts establish coverage. A reference with no countable statements is flagged as an invalid baseline input rather than silently assigned a score. The declaration-counting rule is fixed in the pilot. Each probabilistic condition must produce the same six-category output format. Converting a historical scalar judge into a probability forecast is a new baseline condition and must be documented. Matched artifacts and rating assignments reduce unnecessary comparison noise. Unequal model calls, tokens, and runtime must be reported rather than hidden as equivalent budgets.

The constant forecast tests whether input-dependent predictions add information beyond the training rating frequencies. The count baseline tests whether a cheap structural signal explains the apparent gain. The fixed-rubric condition tests the benefit of fitting or selection, while the correlation-selected condition tests the choice of selection objective. For the last contrast, both selection procedures must search the same predefined candidate set with the same information access and computational budget. Otherwise, an observed difference could reflect search resources or a different model family rather than the objective.

The pilot must set a minimum meaningful improvement, a tolerable distribution error, repeat-instability and failure limits, and the corresponding interval rule. If a claim requires success on all four configurations, inference must account for those four simultaneous claims. These thresholds are currently unset; a statistically nonzero improvement alone is not the proposed definition of practical success. Later changes after test inspection require fresh evidence or an explicitly exploratory label.

Statistic	Estimate	95% interval
Spearman ρ	0.826	[0.316, 0.966]
Kendall τ_b	0.698	[0.216, 0.913]
Pearson r	0.858	[0.707, 0.941]
ICC(A,1)	0.727	[0.391, 0.890]

Table 3: Historical selected ensemble: 30 artifacts from ten test tasks. Intervals are percentile task bootstraps with 10,000 requested draws. ICC(A,1) is single-score absolute-agreement intraclass correlation. The split had prior exposure; these are not fresh-study results.

6 Historical VeriBench Evidence

The historical archive contains several distinct analyses that must not be pooled or described as one original result. An early analysis reports Pearson correlation 0.703349 on 72 scored artifacts from a 75-artifact, 25-task pool. It uses a GPT-5.3-Codex judge with an isotonic score mapping fitted on training folds, against the mean of one expert's two earlier-prompt passes. Its Spearman and Kendall values are 0.321143 and 0.265060. These are two-fold task-split validation results without a separate test counterpart; the aggregate receipt does not supply confidence intervals.

A later selected ensemble, source experiment 100, uses GPT-6 Sol with high reasoning effort and Claude Opus 5.5 with maximum compute effort through the historical Vals model gateway. For each member it takes the median of three calls, then averages the two medians: six calls per completed scalar score. It uses human rating instructions, training-derived conventions, and training examples. The 25 tasks are split 10/5/10 into training, validation, and test, retaining each task's three candidates together. The human target is the mean of available ratings from a three-expert panel, divided by five. Of the 30 test artifacts, 18 have three expert ratings and 12 have two.

The headline correlations in Table 3 coexist with normalized mean absolute error 0.201111 and mean bias $+0.181111$, with judge mean 0.653333 and human mean 0.472222. Bias is the judge rating minus the expert mean,

so a positive value indicates greater leniency. A separately labeled post hoc task bootstrap of the frozen historical data gives 95% intervals [0.130, 0.273] for absolute error and [0.099, 0.262] for bias. It uses 10,000 draws and reproduces the original point estimates; it is new analysis of old data, not new human evidence. No new significance test is asserted for these quantities.

Excluding two test tasks whose references were placeholders when rated leaves 24 artifacts from eight tasks. On that ordinary-reference subset, Spearman correlation is 0.652 with interval $[-0.166, 0.916]$, and Pearson correlation is 0.835 with interval [0.454, 0.958]. The mean bias rises to $+0.205556$, from judge mean 0.566667 and human mean 0.361111. The subset correlation intervals use 1,998 nondegenerate draws out of 2,000 requested task bootstraps. Reference quality therefore changes both the interpretation and the uncertainty of the headline association.

The reported mean repeat standard deviation, 0.020488, also needs its original unit. It is computed from three per-pass averages, each combining one call from each model. It is not the standard deviation of repeated executions of the complete six-call median-based ensemble. The archive supplies no interval for that repeat statistic.

For context, the archived leave-one-expert-out comparison reports Spearman correlation 0.764 over all 75 artifacts and 0.747 over the 45 validation-plus-test artifacts of experiment 100. These compare one expert with the other available experts and have no uncertainty intervals in the imported aggregate. Their task sets and targets differ from the 30-artifact test comparison in Table 3; they cannot establish that the selected judge exceeds human agreement on a matched test set.

Several limitations prevent clean transfer claims. Candidate-generation records trace 74 of 75 files to three single-call, zero-shot samples per task from Claude 3.7 Sonnet at temperature 0.7, with one source receipt missing. These are repeated samples of one older model, not three different coding agents. The source receipt discloses prior exposure to the test ratings in earlier experiments and an initial descriptive audit; the exact total number of exposed judge versions and resulting selection inflation are not established by the inspected records. Nine references changed statement content between rating and earlier judging, although experiment 100 restored rating-era inputs. The reconstruction remains ambiguous for five tasks whose references changed during the rating window; six test artifacts from two tasks carry that ambiguity flag. The source appendix also reports three test items whose rating sheet linked a different candidate artifact. Excluding either group raises test agreement in the source analysis, but does not repair the original evidence or justify replacing its full denominator. Neither adding raters to those files nor narrowing their intervals would erase these provenance limits.

Finally, a separate five-fold automated recipe, source experiment 104, reports tie-preserving Spearman correlation 0.603 [0.287, 0.836] over all 75 artifacts and 25 tasks, with Kendall 0.498 and Pearson 0.621. Its original floating-point calculation gave Spearman 0.601533; the published addendum preserves mathematical ties before ranking. This evaluates fold-specific recipe fits, not the frozen selected ensemble. The archive retains both versions and their labels. All of this historical evidence concerns scalar LLM judges; none measures the proposed probability forecasts or validates a prover judge.

7 Semantic Controls and Reporting

Agreement with experts using a faulty reference can reproduce the same error rather than measure coverage correctly. We therefore accompany the random audit with controls whose semantic relationship is checked independently wherever possible. These controls do not receive the random audit's sampling interpretation, and their counts and selection process are reported separately.

Removing a necessary obligation should reduce coverage, provided no remaining statement already implies it. Narrowing a precondition should reduce coverage when it excludes inputs required by the task. Equivalent formulations should receive similar ratings, while adding irrelevant or vacuous statements should not increase meaningful coverage. Changing proof bodies without changing statements should preserve a statement-coverage rating. An incomplete or false reference should trigger discrepancy reporting where the evidence supports it. The exact tolerances and control counts are pilot decisions, not assumed measured properties of the judge.

For equivalence controls, compilation of two programs is insufficient. The relevant implications must preserve domains, assumptions, and obligation weights. For negative controls, a counterexample should establish that the removed obligation is not already entailed. AI can assist in constructing these certificates, but checker acceptance and the assumptions of the certificate must be recorded. Unresolved cases remain unresolved rather than being converted into negative examples by a timeout.

The reference-anchoring study is a separate controlled comparison within this research program. It varies reference representation and proof visibility while holding the intended obligations and candidate artifact fixed. Its labels, estimand, and pilot counts must not be substituted for those of the main conditional-rating study. Likewise, the human-rating study does not change an existing benchmark's prover-based scoring implementation.

7.1 What the final report must contain

The primary table will report RPS, paired baseline differences, and confidence intervals for all 896 audit tasks, the 128 seen tasks with fresh outputs, the 768 held-out tasks, and each solver configuration. The four excluded-solver fits have their own transfer table and denominators. Distribution-error diagnostics on the dense subset, means, signed bias, absolute error, Pearson correlation, Spearman correlation, and Kendall correlation supplement these results. Human–human agreement is computed on matched artifacts, and the prediction mean is not confused with the full probability vector.

A separate repeatability panel reports variation across repeated probability vectors and expected scores, the number

of full judge executions, and the aggregation policy. A stable but wrong judge can pass a repeatability check, while a correct prediction of human disagreement can intentionally be broad. The paper must keep these statements separate. Missing and failed calls appear with their original denominator; because RPS is bounded, worst- and best-case completion bounds can show whether missingness could reverse a conclusion.

Public evidence consists of the protocol, configurations, input manifests where distributable, analysis code, and authorized aggregate results. Participant identities, contact information, consent, individual rating sheets, and sealed test ratings remain in restricted storage. Opaque analysis identifiers preserve the assignment structure without exposing identities. Any eventual release of individual ratings requires the corresponding consent and release decision. The historical import preserves immutable aggregate receipts and hashes, with post hoc calculations labeled separately.

8 Discussion and Current Status

The proposed objective makes a precise claim possible: under the stated input conditions and human population, the judge predicts a conditional ordinal rating distribution with an estimated error and uncertainty. It does not equate agreement with semantic truth, prove that every artifact is scored correctly, or establish transfer to all future agents. Controlled semantic checks and prover evidence provide complementary evidence about the construct being measured.

The 896-task, four-configuration, 24-rater allocation is a defensible starting design, not a universal sample-size theorem. The broad task audit addresses the narrow historical task sample; distinct solver fits address a missing transfer question; separate raters and additional ratings address different kinds of human uncertainty. Pilot estimates may show that the limiting resource is expert time, that the rater pool is too narrow, or that a desired precision is infeasible. Such findings require a declared revision of the claim or allocation rather than presenting a smaller completed subset as the original design.

Current deliverables are the methodology, machine-readable collection budget, reproducibility checks, historical aggregate archive, and this paper draft. No fresh rating round, probabilistic judge selection, or final audit result is reported here. Exact solver pins, practical success tolerances, repeat counts, and the final sparse-assignment inference procedure remain to be frozen after the pilot and before final test inspection. For evaluation of AI scientists, the broader lesson is operational: an instrument should be evaluated against an explicit target with uncertainty and transfer claims that correspond to the actual data, rather than endorsed by a single favorable correlation.

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